Problem Set 1
LUISS Advanced Macroeconomics
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Exercise 1: Business Cycle Data

I use quarterly real GDP and quarterly real personal consumption expenditure for the United States, together with quarterly real GDP and quarterly real private consumption expenditure for Canada. Since the Canadian consumption series starts later than the U.S. series, I truncate the U.S. sample so that all four series share the same common time span. I then take logs of each series and extract the cyclical component using the Hodrick–Prescott filter with smoothing parameter $\lambda = 1600$, appropriate for quarterly data. The cyclical component is the deviation of the logged series from its estimated trend.

1.1 Detrended series

Figures 1 and 2 plot the detrended log GDP and detrended log consumption series for the United States and Canada, respectively.

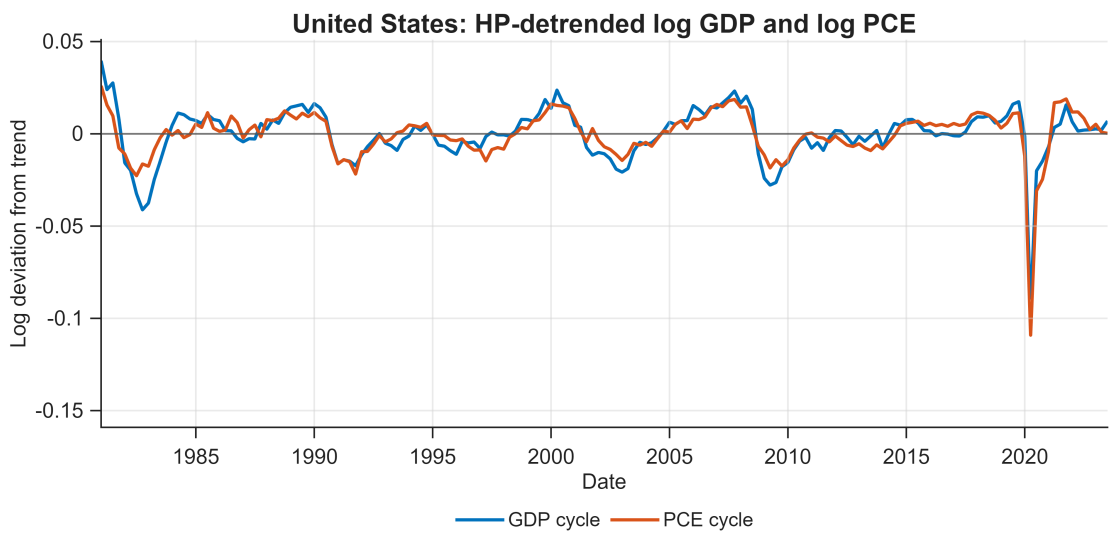


Figure 1: United States: HP-detrended log GDP and log PCE

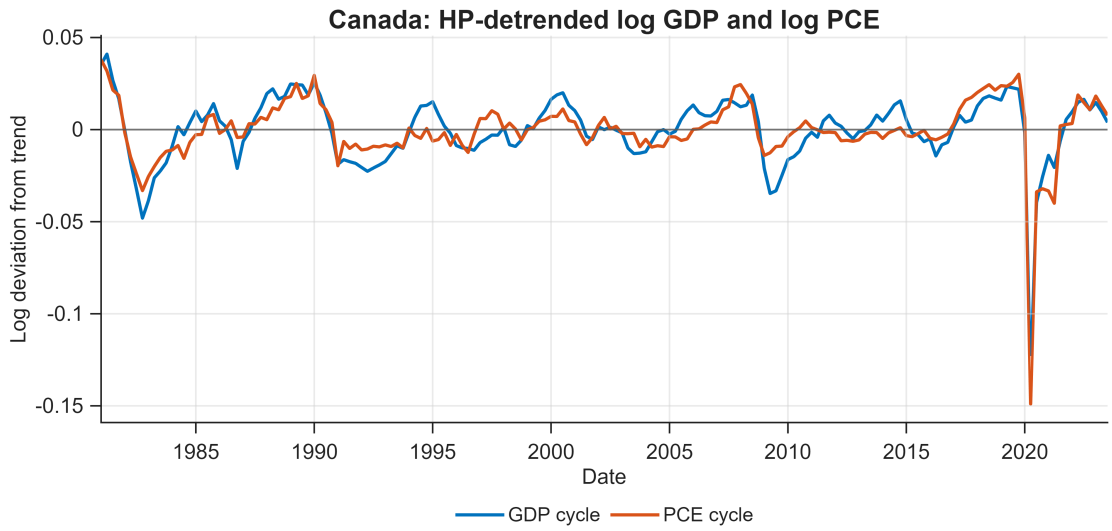


Figure 2: Canada: HP-detrended log GDP and log PCE

In both countries, the cyclical components of GDP and consumption move closely together over time. The figures also show a particularly sharp contraction in 2020, followed by a rapid rebound.

1.2 Variances and correlations

Table 1 reports the variance of detrended GDP and consumption for the United States and Canada, together with the requested correlations.

Table 1: Business-cycle moments

Statistic	United States	Canada
Variance of GDP cycle ($\times 10^4$)	1.9368	3.0865
Variance of PCE cycle ($\times 10^4$)	1.6508	2.9399
Correlation between GDP and PCE cycles	0.8808	0.8736
Correlation with U.S. GDP cycle: GDP	—	0.8668
Correlation with U.S. GDP cycle: PCE	—	0.7739

1.3 Comments

The results reveal several clear patterns. In the United States, detrended consumption is less volatile than detrended GDP, as the variance of the consumption cycle is lower than that of the output cycle. This is consistent with the idea of consumption smoothing: households tend to adjust consumption less than income or output in response to cyclical fluctuations. The two U.S. cyclical series also display strong positive comovement, with a correlation of 0.881. Canada appears more volatile than the United States, since both the variance of detrended GDP and the variance of detrended consumption are higher over the common sample. At the same time, the Canadian cycle is strongly synchronized with that of the United States: the correlation between Canadian and U.S. detrended GDP is 0.867, while Canadian detrended consumption is also positively correlated with U.S. detrended GDP, at 0.774. Overall, the evidence suggests that consumption is smoother than output, that output and consumption comove strongly within each country, and that Canadian macroeconomic fluctuations are closely aligned with the U.S. business cycle.

Exercise 2: Local Projections and Macroeconomic Shocks

2.1 Replication

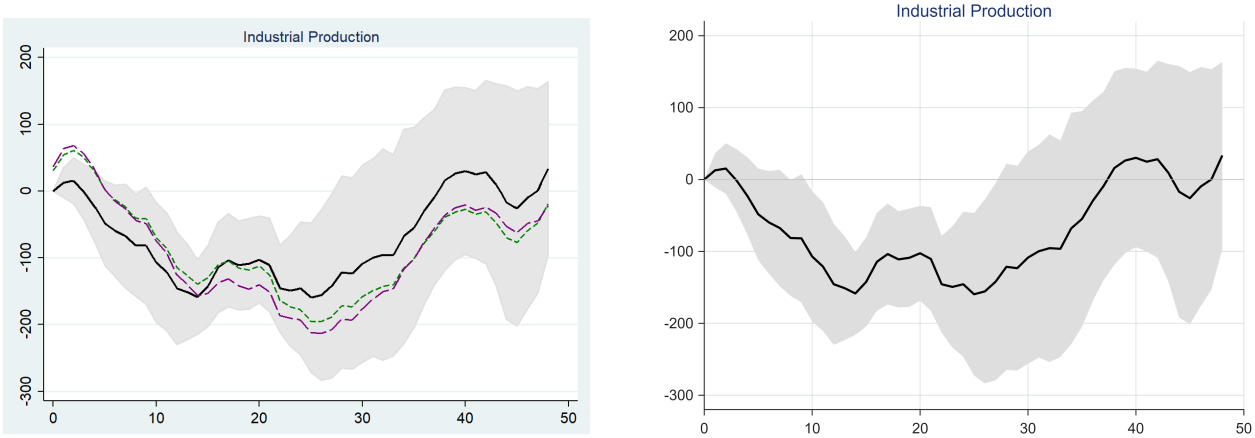
In this part, I replicate the local-projection response of industrial production to a Romer–Romer monetary policy shock using the replication files accompanying Ramey (2016), Chapter 2. The empirical question is whether a contractionary monetary policy shock leads to a decline in real activity, as measured by industrial production. I focus on industrial production because it is one of the main real-side variables reported in the chapter.

The shock series is the original Romer–Romer monetary shock, recovered from the cumulative series in the replication files. The outcome variable is industrial production. I use the monthly sample from 1969m3 to 1996m12 and estimate Jordà local projections up to a horizon of 48 months. The specification follows the baseline recursive version in the replication files. More precisely, for each horizon h, I estimate a regression of the form

$$y_{t+h} = \alpha_h + \beta_h \, \text{shock}_t + \Gamma_h' X_t + u_{t+h},$$

where y_{t+h} is industrial production at horizon h, shock_t is the contemporaneous Romer–Romer shock, and X_t includes contemporaneous and lagged values of industrial production, unemployment, the price level, commodity prices, and lagged values of the federal funds rate. I use two lags and compute 90% confidence intervals using Newey–West standard errors with horizon-dependent bandwidth.

Figure 3 compares the original industrial production panel from Figure 2B with my replication. The replicated impulse response closely matches the original figure in terms of overall shape, timing of the trough, and subsequent recovery. In that sense, the chart can be viewed as an accurate replication of the industrial production response reported in the chapter.



(a) Original figure

(b) MATLAB replication

Figure 3: United States: Industrial production response to an original Romer–Romer monetary shock

The replicated impulse response indicates that a contractionary monetary policy shock leads to a persistent decline in industrial production, with the strongest effect appearing only after several months. This delayed and hump-shaped response is economically plausible and consistent with the idea that monetary policy affects real activity with lags rather than immediately on impact.

2.2–2.4 Alternative shock, estimation, and discussion

For the second part, I examine whether large oil price run-ups are followed by an increase in the U.S. consumer price level. To do so, I use a net oil price increase measure based on Hamilton (2003) as the shock and estimate the response of headline CPI using local projections.

This shock–outcome pair is well suited to the question. On the shock side, using raw monthly oil price changes would treat all oil price movements symmetrically, even though small reversals and short-lived fluctuations are unlikely to have the same macroeconomic relevance as sharp oil price run-ups. Hamilton's measure is designed precisely to isolate episodes in which the oil price exceeds its recent peak, thereby focusing on large positive oil price innovations rather than ordinary month-to-month noise. On the outcome side, headline CPI is the natural price variable because oil shocks should affect the aggregate consumer price level most directly through energy-related components and, more broadly, through higher transportation and production costs. The empirical question is therefore whether these unusually large oil price increases are followed by a persistent rise in the overall price level.

I construct a monthly U.S. dataset from FRED. The price variable is the Consumer Price Index for All Urban Consumers: All Items (`CPIAUCSL`). The oil price series is Crude Oil Prices: West Texas Intermediate (WTI) – Cushing, Oklahoma (`MCOILWTICO`). The control variables are the Unemployment Rate (`UNRATE`), the Effective Federal Funds Rate (`FEDFUNDS`), and the 1-Year Treasury Constant Maturity Yield (`GS1`). From the WTI series, I construct the net oil price increase measure using a 36-month lookback window,

$$HAM36_t = \max\left\{0,\ \log P_t - \max(\log P_{t-1}, \ldots, \log P_{t-36})\right\}.$$

In the estimation I use $100 \times HAM36_t$, while the outcome variable is $100 \times \log(\text{CPI})$, so the responses are measured in log points.

I estimate local projections for horizons $h = 0, \ldots, 24$ months and report 90% confidence intervals. The specification includes the contemporaneous oil shock and two lags, the contemporaneous outcome and two lags, the contemporaneous unemployment rate and two lags, two lags of the federal funds rate, and three lags of the 1-year Treasury rate. Standard errors are computed using Newey–West HAC with bandwidth $h + 1$. Figure 4 shows the estimated impulse response of the CPI level to the Hamilton oil shock.

The estimated response is positive at essentially all horizons. Following the shock, the CPI level rises on impact, reaches about 0.04–0.05 log points after a few months, and remains above zero throughout the two-year horizon. This pattern is consistent with the idea that large oil price run-ups pass through, at least to some extent, into headline consumer prices.

At the same time, the figure should be interpreted with caution. The confidence bands widen substantially at medium and longer horizons, so the estimates provide clearer evidence on the sign of the response than on its precise magnitude or persistence. In addition, because the exercise is reduced-form, the estimated response should not be interpreted as isolating a purely exogenous structural oil supply shock. Rather, it shows that episodes in which oil prices rise above their recent peak are followed by higher headline CPI in the data. More generally, this is a deliberately parsimonious specification, so the results should be viewed as suggestive rather than definitive. A richer specification could allow for additional controls and alternative channels through which oil price movements and inflation co-move. Overall, the results point to economically plausible positive pass-through from major oil price run-ups to the aggregate price level, while leaving the exact magnitude and persistence of the effect somewhat uncertain.

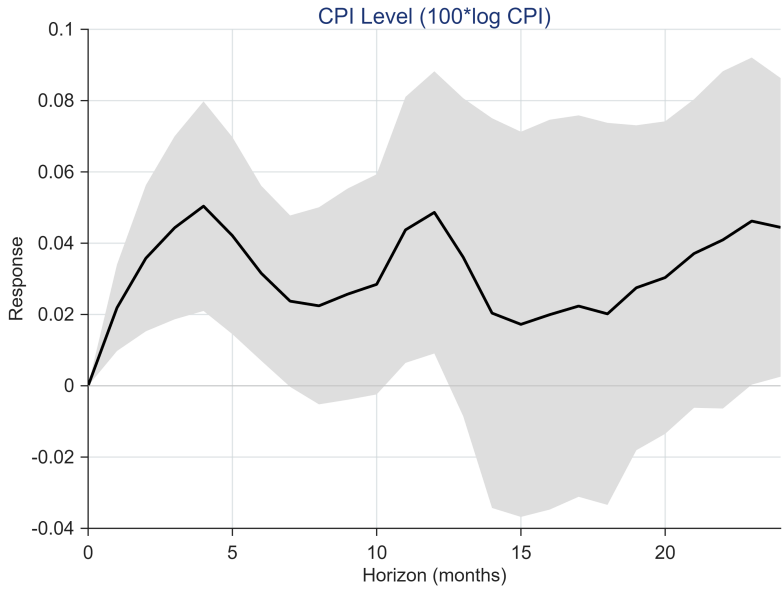


Figure 4: United States: CPI level response to a Hamilton oil shock

References

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